Raghav Thakar

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EDUCATION

Oregon State University — Corvallis, OR, USA
Ph.D. in Robotics, CGPA: 3.73/4.00, Advisor: Dr Kagan Tumer — September 2023 – June 2027 (Expected)
- Prospective dissertation theme: Learning-based control of adaptive autonomous agents in environments with multiple objectives and sparse feedback.
- Keywords: Sample-efficient learning, multi-objective reinforcement learning, multi-agent reinforcement learning, offline reinforcement learning, reward shaping, evolutionary algorithms, distributed intelligence.

Manipal Institute of Technology — Manipal, KA, India
Bachelor of Technology (B.Tech.) in Computer Science and Engineering, CGPA: 9.00/10 — July 2019 – April 2023
- Minor Specialisation in Computational Mathematics

EXPERIENCE

Autonomy Intern – Autonomous Behaviours Team — June 2026 – September 2026
Skydio Inc. — San Mateo (San Francisco Bay Area), CA, USA
- Architected a distributed coordination system for production Skydio X10 drones, enabling fleet-wide, sub-second (profiled at 250ms) synchronisation to reliably perform multi-drone missions in critical domains.
- Leveraged core distributed computing concepts to implement a new cloud service that empowers multiple independently-deployed cloud frontends to agree on a single, shared synchronisation primitive for the drones in their scope.
- Integrated and deployed this system with the complete Skydio autonomy and software stack, spanning the Skydio X10 drone, the customer-facing Remote Flight Deck (RFD) application, and production cloud software.
- Performed testing, validation, and profiling with a real-life synchronised multi-drone mission, namely a 14-drone air show executing a bespoke choreography, time-synchronised to music.

Research Intern — January 2023 – April 2023
MOON Lab, IISER Bhopal. Advisor: Dr Sujit PB — Bhopal, MP, India
- Developed a COLREGS-compliant multi-agent path planning and collision avoidance strategy for Unmanned Surface Vehicles (USVs) based on the Conflict-Based Search algorithm.
- Deployed the navigation pipeline on an ArduPilot/Pixhawk stack, implementing custom waypoint-generation logic to translate global plans into executable autonomous trajectories.
- Architected a mixed-reality testbed that integrated physical hardware with virtual agents via a UDP communication system (Ubiquiti Rocket M5) and performed real-time multi-agent field trials.
- **In parallel:** Participated in the 2023 MBZIRC Grand Maritime Challenge; implemented a multi-agent coverage algorithm for autonomous patrolling by drones and finished 12th out of 300+ teams globally. [Team Luna]

MITACS Globalink Research Intern — May 2022 – July 2022
MARS Lab, Ontario Tech University. Advisor: Dr Scott Nokleby — Oshawa, ON, Canada
- Designed and implemented a hybrid navigation framework for an autonomous wheelchair, combining RRT* for global planning with Artificial Potential Fields for local obstacle avoidance.
- Integrated an Intel RealSense depth camera and 2D LiDAR for real-time mapping, obstacle detection, and robot localisation.
- Validated the full autonomy stack on a TurtleBot platform in cluttered indoor environments.

PUBLICATIONS

- J. Santos, A. A. Aydeniz, **Raghav Thakar**, Kagan Tumer. *Entropy-Augmented Multi-Objective Policy Optimization in Multiagent Systems*. IJCAI-ECAI 2026 - MODeM Workshop. [Paper]
- **Raghav Thakar**, J. Santos, A. A. Aydeniz, Gaurav Dixit, Kagan Tumer. *Reward Sparsity in Multi-Objective Reinforcement Learning: A Taxonomy and Key Challenges*. IJCAI-ECAI 2026 - MODeM Workshop. [Paper]
- **Raghav Thakar**, Gaurav Dixit, Kagan Tumer. *Post Hoc Extraction of Pareto Fronts for Continuous Control*. IJCAI-ECAI 2026. [Paper]
- **Raghav Thakar**, Kagan Tumer. *Cultivating Divergent Multi-Objective Expertise in Multiagent Systems via Expert Ensembles*. AAMAS 2026 - ALA Workshop. [Paper]
- **Raghav Thakar**, Gaurav Dixit, Siddarth Viswanathan, Kagan Tumer. *Multiagent Credit Assignment for Multi-Objective Coordination*. GECCO 2025. [Paper]
- Everardo Gonzalez, **Raghav Thakar**, Kagan Tumer. *Indirect Credit Without a Heuristic for Coevolving Agents*. GECCO 2025 - Late-Breaking Abstract. [Paper]
- **Raghav Thakar**, Kagan Tumer. *The Specificity vs. Expense Trade-Off of Multiagent Credit*. GECCO 2025 - Student Workshop. [Paper]
- **Raghav Thakar**, Gaurav Dixit, Kagan Tumer. *Multi-Objective Credit Assignment for Multiagent Systems*. ECAI 2024 - MODeM Workshop. [Paper]

PROJECTS

Evolutionary Reinforcement Learning for Multi-Objective Multi-Agent Coordination [Project page] 2024
- Combined evolutionary optimisation with multi-agent reinforcement learning to learn a Pareto front of multi-agent joint-policies that span trade-offs across conflicting coordination objectives.

Closed-Loop Planning with Large Language & Vision Models [Project page] 2025
- Engineered a system to integrate a Large Language Model (LLM) and Vision-Language Model (VLM) into an agent's planning stack, enabling spatial reasoning and iterative performance improvement via multimodal feedback.

Optimal Transitions to Distribute Heterogeneous Agents Across Tasks [Project Page] 2022
- Formulated and solved an Integer Programming problem to compute optimal transition probabilities within a Markov chain, ensuring steady-state task satisfaction for distributed multi-agent networks.

Decentralised Multi-Agent Formation Control [Project Page] 2022
- Designed a distributed coordination system by adapting the Token Ring algorithm to manage communication and spatial alignment across an autonomous multi-agent team of drones.

AWARDS AND RECOGNITION

Best Poster and Presentation, OSU AI Week 2025
- Awarded at the AWS-sponsored 'Lightning Talk' event.

MITACS Globalink Research Internship 2022
- Awarded a fully funded scholarship by MITACS Canada for a summer research internship on autonomous wheelchair navigation for children with motor disorders at Ontario Tech University.

Development Grant, Swadeshi Microprocessor Challenge 2021
- Received a grant of Rs 100,000 (approx $1500) from the Govt. of India to develop a novel hardware prototype using the then under-development indigenous 'Shakti' microprocessor; finished in the top 100 of 3,000+ teams nationally.

TECHNICAL SKILLS

- **Languages & Libraries:** Python (PyTorch, JAX, NumPy, Pandas), C++, Bash, LaTeX
- **AI & Machine Learning:** Reinforcement Learning (RLHF, Multi-Agent, Multi-Objective, Offline), Reward Modelling/Shaping & Credit Assignment, Large Language Models (LLMs), Vision-Language Models (VLMs), Multi-Objective Optimisation, Evolutionary Algorithms
- **Compute & Infrastructure:** Large-Scale HPC Experiments, Distributed Training, Slurm, Experiment Tracking (WandB/Weights & Biases), Git, Linux

GRADUATE COURSEWORK

ROB 537: Learning-Based Control, ROB 538: Multiagent Systems, AI 533: Intelligent Agents and Decision-Making, CS 514: Algorithms, ROB 599: Robot Software Frameworks, ROB 571: Robotics and Society, ME 517: Optimisation in Design.